

On Computing Certain Integrals, and Implications for Symbolic Algebraic Manipulation

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ABSTRACT

Overcoming numerical difficulties in the calculation of certain integrals in an atomic physics computation provides an interesting application of symbolic algebraic manipulation. Complications in doing the symbolic calculation have implications for the design and use of symbolic algebra systems.

KEY WORDS AND PHRASES: Symbolic Algebraic Manipulation

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THE SUCCESS STORY

In the investigation of the possibility that there is a stable bound state of a positron with a helium atom [4], a standard variational procedure was followed: a trial wave function, depending on certain parameters, was applied to the energy functional of the system, and the parameters varied to find the minimum energy. Fixing one of the parameters at zero corresponded to the unbound situation, and the energy minimum with this parameter at zero is that of the helium atom alone, and was well known. If any lower values existed for other values of the parameters, a bound state was indicated, and values at the minimum for specific terms in the energy, the energy itself, and the parameters (all of which were physically meaningful) would be of interest. The minimization process is numerical, and the answers we want are numbers; thus, we could assert that the problem is numerical.

Computing the energy required, among other things, computing the value of many integrals of the form

$$I(r,s,t) = \int_0^\infty \frac{dk}{(k^2+a^2)^r (k^2+b^2)^s (k^2+c^2)^t} \quad (1)$$

where a , b , and c depend on the trial set of parameters and many sets of r , s , and t were involved. How should these integrals be computed?

One possibility is numerical quadrature, but it is an unsatisfactory method because of the non-polynomial behaviour of the integrand even after transformation of variables and judicious choice of weight functions, so that quadrature rules using many function evaluations are needed to achieve adequate accuracy.

An alternative approach is to observe that using the identity

$$\frac{1}{(k^2+a^2)(k^2+b^2)} = \frac{1}{b^2-a^2} \left(\frac{1}{k^2+a^2} - \frac{1}{k^2+b^2} \right) \quad (2)$$

we can obtain the recurrences

$$I(r,s,t) = (I(r,s-1,t) - I(r-1,s,t))/(b^2-a^2)$$

and

$$I(r,s,t) = (I(r,s,t-1) - I(r,s-1,t))/(c^2-b^2) \quad (3)$$

where the degrees in the denominator are dropped by one, so that starting from

$$I(r,0,0) = \frac{(2r-2)!}{2^{2r-2} [(r-1)!]^2} \frac{\pi}{2} \frac{1}{a^{2r-1}} \quad (4)$$

and similar expressions for $I(0,s,0)$ and $I(0,0,t)$ with specific values a , b , and c , the integrals can be calculated in turn numerically. Unfortunately, this recurrence method can lead to disastrous loss of accuracy due to cancellation in the numerator and denominator if a , b , or c approach each other. Although this problem does not occur at the minimum that was found, it did occur during the minimization process.

A third means of attack must therefore be considered - the analytic form of these integrals must be found, and numerical substitution must be made into these forms. Various methods are possible for obtaining the analytic form of these integrals, including the obvious suggestion that the recurrences (3) be marched symbolically. Of course, if we are not careful, the numerical instability of the recurrence relation will reappear as severe cancellation when substituting into the analytic form, but if we use a GCD computation to reduce the analytic form to its lowest terms (clearly, from the recurrences, $I(r,s,t)$ is a rational algebraic in π and a , b , c) a most remarkable phenomenon occurs: the numerator and denominator of $I(r,s,t)$ each reduce to polynomials in a , b , c with only positive coefficients! Consequently, no cancellation occurs in substitution and the evaluation is extremely stable numerically. In principle, marching these recurrences symbolically could be carried out by hand, but in practice the analytic forms of the integrals become unpleasantly

complicated, eg.

$$\begin{aligned}
 I(3,2,1) = & \pi \{ 8a^6b + 4a^6c + 32a^5b^2 + 40a^5bc + 12a^5c^2 + 48a^4b^3 \\
 & + 108a^4b^2c + 72a^4bc^2 + 12a^4c^3 + 32a^3b^4 + 112a^3b^3c \\
 & + 132a^3b^2c^2 + 56a^3bc^3 + 4a^3c^4 + 8a^2b^5 + 52a^2b^4c \\
 & + 96a^2b^3c^2 + 68a^2b^2c^3 + 16a^2bc^4 + 9ab^5c + 30ab^4c^2 \\
 & + 33ab^3c^3 + 12ab^2c^4 + 3b^5c^2 + 6b^4c^3 + 3b^3c^4 \} \\
 & \{ 16a^{12}b^5c + 32a^{12}b^4c^2 + 16a^{12}b^3c^3 + 64a^{11}b^6c + 176a^{11}b^5c^2 \\
 & + 160a^{11}b^4c^3 + 48a^{11}b^3c^4 + 96a^{10}b^7c + 384a^{10}b^6c^2 \\
 & + 528a^{10}b^5c^3 + 288a^{10}b^4c^4 + 48a^{10}b^3c^5 + 64a^9b^8c \\
 & + 416a^9b^7c^2 + 832a^9b^6c^3 + 688a^9b^5c^4 + 224a^9b^4c^5 \\
 & + 16a^9b^3c^6 + 16a^8b^9c + 224a^8b^8c^2 + 688a^8b^7c^3 \\
 & + 832a^8b^6c^4 + 416a^8b^5c^5 + 64a^8b^4c^6 + 48a^7b^9c^2 + 288a^7b^8c^3 \\
 & + 528a^7b^7c^4 + 384a^7b^6c^5 + 96a^7b^5c^6 + 48a^6b^9c^3 + 160a^6b^8c^4 \\
 & + 176a^6b^7c^5 + 64a^6b^6c^6 + 16a^5b^9c^4 + 32a^5b^8c^5 + 16a^5b^7c^6 \}
 \end{aligned} \tag{5}$$

The manipulation of such expressions is exceedingly tiresome; moreover, it is hard to avoid mistakes. Our purely numerical problem has developed a requirement for considerable algebraic computation, and the use of an algebraic manipulation system is desirable. Once the analytic forms are found, of course, we will return to purely numerical computation, so either the algebraic manipulation system should provide floating point arithmetic facilities or its output should be compatible with the input source text form of some language for numerical computation. (One possible difficulty in using algebraic manipulation system output as source text for a numerical computation language is that the size of expressions may well exceed that allowed for the parse tree in the translator for that language).

We used ALTRAN [3], which provides both of the above features. A program which does not include the numerical part may be found in the Appendix. The program implements the recurrence relations (3) directly, but obtains the values of $I(r,0,0)$ from the recurrence

$$I(1,0,0) = \frac{\pi}{2a} ; \quad I(r,0,0) = \left(\frac{r-3}{r-1} \right) \frac{I(r-1,0,0)}{a^2} \tag{6}$$

Since permutations of the values of indices r , s , and t merely produce the same analytic form with a , b and c similarly permuted, clearly each combination of values need only be used once, and we can take $r \geq s \geq t$. Each step in the recurrence will still involve only integrals already computed, except for the computation of $I(r,r,t)$ and $I(r,s,s)$ which require $I(r-1, r, t)$ and $I(r, s-1, s)$ respectively. By interchanging a and b in $I(r, r-1, t)$,

which is computed, $I(r-1, r, t)$ can be obtained, and similarly by interchanging b and c in $I(r, s-1, s)$, which is computed, $I(r, s-1, s)$ can be obtained.

THE OTHER STORY

When the program is run, the required results are obtained, but at exorbitant cost and with intermediate results having huge coefficients and ridiculous numbers of terms, that is, the program exhibits classical intermediate expression swell. The difficulty is not hard to trace; we are being "GCD'ed" to death. ALTRAN uses GCD's in every operation to ensure that numerators and denominators of the rational algebraic expressions it manipulates are relatively prime. (As mentioned earlier, apart from the usual reason of desiring uniqueness of representation, this property is specifically necessary for the usefulness of the analytic forms for our integrals). This is extremely expensive in the current version of ALTRAN, which uses Collins' subresultant PRS algorithm for computing GCD's [2, 1]. The subtractions are expensive because the denominators have large GCD's, moreover, the process of verifying that $I(r-1, r, t)$ and $I(r, s-1, s)$ obtained by swapping variables are already in the correct form (which of course they are since $I(r, r-1, t)$ and $I(r, s, s-1)$ were) is particularly expensive: it appears that the subresultant PRS's obtained are abnormal [1].

Before concluding that we have merely found an example where the subresultant PRS is unsatisfactory and the more powerful homeomorphism methods [1] should be used, we should stop and note that the problem is really one of our own making. Examining

the recurrences readily establishes that the denominators for any of these integrals must actually be products of powers of a , b , c , $(a+b)$, $(a+c)$, and $(b+c)$. The current version of ALTRAN, like many other systems, forces the numerator and denominator to be completely expanded, but if instead a factored representation of each could be maintained (as indeed later versions of ALTRAN will do) then the expensive GCD calculations would be replaced by the very inexpensive divide test computation that often precedes entry to a complete GCD routine of any type. This would happen because the recurrences (including the swapping of variables) actually build up the integrals in factored form. An added benefit would be that the factored form of the answers would be useful in optimizing the numerical evaluation of the integrals.

LESSONS

This example has illustrated the following lessons for potential users of an algebraic manipulation system:

- U1) Even in a purely numerical problem, doing some stage by algebraic manipulation can be useful, particularly in overcoming numerical instability.
- U2) Assessing which of several algebraic computations is most economical in today's algebraic manipulation systems is not obvious. For example, substitution of new parameter values into an expression already derived can be more expensive than deriving it again, using the new values of the parameters.

The example has also illustrated the following lessons for implementors of algebraic manipulation systems:

- I1) Algebraic manipulation may be desirable for part of a computation which is primarily numerical; therefore, some means of performing numerical computation with formulae obtained from the algebraic stage is required. Hand transcription from the output format of the algebra system to the input source language format of some numeric processing language is not satisfactory.
- I2) A factored representation for rational algebraic expressions can often exploit factoring information available from the construction of the expressions to make significant economies in computation as well as to provide answers in a form the user may prefer.
- I3) The strategy [5, p.291] of always doing GCD calculations to ensure that rational algebraic expressions are represented in canonical form by numerators and denominators that are relatively prime is questionable. Unnecessary and expensive computations occur when external information indicating that the canonical form already exists is ignored. Moreover, even when final results are required in canonical form, it is not obvious that each intermediate result should be in that form. Neither is this strategy always the cheapest way to get the final result in canonical form, nor is it always the best way to control intermediate expression swell. Unless we find that in practice factored representations and homeomorphism GCD algorithms provide the significant reduction

in the cost of an average GCD computation that they promise, users will want calls for GCD computations to be explicit.

- I4) The general problem of substitution of algebraic expressions for the indeterminates in some other algebraic form is an expensive and not well understood problem, but there are important special cases, such as the variable swap done here, for which it is imperative to be more efficient than the general case can be, and for which efficient methods can be constructed. Without suggesting the ridiculous growth of systems that would result from attempting to exploit every conceivable special case, we would recommend that some cases be treated specially.

EPILOGUE

The referee has suggested that it is only fair to describe how the problem really should be solved. One possibility would be to use a difficult kind of recurrence [4],

$$I(r+1, s, t) = -\frac{1}{2ar} \frac{\partial}{\partial a} I(r, s, t) \quad (7)$$

and the similar formulae differentiating with respect to b and c , starting from $I(1, 0, 0)$, $I(1, 1, 0)$ and $I(1, 1, 1)$. This, unfortunately, doesn't even work as well as the previous method.

A more successful method is to examine the recurrences (3) and (4) and observe that if we write

$$I(r, 0, 0) = -\frac{\pi}{2} \frac{P_1(r)}{a^{2r-1}} \quad (8)$$

$$I(r, s, 0) = \frac{\pi}{2} \frac{P_2(r, s)}{a^{2r-1} b^{2s-1} (a+b)^{r+s-1}} \quad (8')$$

$$I(r, s, t) = \frac{\pi}{2} \frac{P_3(r, s, t)}{a^{2r-1} b^{2s-1} c^{2t-1} (a+b)^{r+s-1} (a+c)^{r+t-1} (b+c)^{s+t-1}} \quad (8'')$$

then for r, s , and t at least one, P_1 is simply a rational number, P_2 is a polynomial in a and b with rational coefficients, and P_3 is a polynomial in a , b , and c with rational coefficients. Recurrences for these can be obtained from (3) and (6)

$$P_1(1) = 1; \quad P_1(r) = \frac{(r-3/2)}{(r-1)} P_1(r-1) \quad (9)$$

$$P_2(r, 0) = \frac{(a+b)^{r-1}}{b} P_1(r); \quad (9')$$

$$P_2(r, s) = \frac{(b^2 P_2(r, s-1) - a^2 P_2(r-1, s))}{(b-a)}$$

$$P_3(r, s, 0) = \frac{(a+c)^{r-1} (b+c)^{s-1}}{c} P_2(r, s); \quad (9'')$$

$$P_3(r, s, t) = \frac{(c^2 (c+a) P_3(r, s, t-1) - b^2 (b+a) P_3(r, s-1, t))}{(c-b)}$$

Marching these recurrences is very quick indeed, leading to one final lesson for users

U3) Polynomial calculations are an order of magnitude easier than rational algebraic calculations, and reformulating your problem that way, if possible, can be very worthwhile.

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APPENDIX

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PROCEDURE MAIN
# CERTAIN INTEGRALS USED IN CALCULATING THE BINDING ENERGY OF HYDROGEN-
# LIKE ATOMS SATISFY A RECURRENCE RELATION. IF THE RECURRENCE IS USED
# NUMERICALLY, DISASTEROUS CANCELLATION OCCURS. THIS PROGRAM COMPUTES
# THE INTEGRALS SYMBOLICALLY, WHEREUPON NUMERICAL EVALUATION IS STABLE.
INTEGER LIMIT=5,R,S,T
LONG ALGEBRAIC (A:500,B:500,C:500,PI:4)
LONG ALGEBRAIC ARRAY (1:LIMIT,0:LIMIT,0:LIMIT) I
DO P=1,LIMIT
  IF (R.EQ.1) I(1,0,0)=PI/(2*A)
  ELSE I(R,0,0)=(R-3/2)/(R-1)*I(R-1,0,0)/A**2
  WRITE I(R,0,0)
  DO S=1,R
    IF (S.NE.R) DO; I(R,S,0)=(I(R,S-1,0)-I(R-1,S,0))/(B**2-A**2)
      I(R-1,S,0)=.NULL.; DOEND
    ELSE I(R,R,0)=(I(R,R-1,0)-I(R,R-1,0)(A,B=B,A))/(B**2-A**2)
    WRITE I(R,S,0)
    DO T=1,S
      IF (T.NE.S) DO; I(R,S,T)=(I(R,S,T-1)-I(R,S-1,T))/(C**2-B**2)
        I(R,S-1,T)=.NULL.; DOEND
      ELSE I(R,S,S)=(I(R,S,S-1)-I(R,S,S-1)(B,C=C,B))/(C**2-B**2)
      WRITE I(R,S,T)
    DOEND
  DOEND
DOEND
END

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